Mathematics for Machine Learning

Part I: Mathematical Foundations

Based on Deisenroth, Faisal & Ong (2020)

Outline

Introduction and Motivation

Linear Algebra

Analytic Geometry

Summary

Introduction and Motivation

What Is Machine Learning?

Machine learning is about designing algorithms that automatically extract valuable information from data.

Three concepts sit at the core of any ML system:

Data

The core of ML: numerical representations (vectors) of observations.

Model

A simplified process, similar to the one that generated the data, used to explain/predict it.

Learning

Automatically finding patterns by optimizing the model's parameters w.r.t. a performance measure.

Paraphrasing Mitchell (1997)

A model is said to *learn* from data if its performance on a given task improves after the data is taken into account. The goal is to generalize well to yet-unseen data.

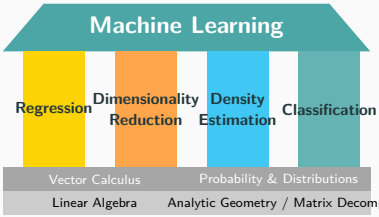
Three Core Ideas of This Book

- We represent data as vectors.
- We choose an appropriate model, using either the probabilistic or the optimization view.
- We learn from data using numerical optimization methods, aiming for good performance on data *not* used for training.

Two ways to read this book

	Bottom-up	**Top-down**
Idea	Build from foundational to advanced concepts	Drill down from practical needs to requirements
Pro	Reader can rely on previously learned concepts	Motivation is always clear
Con	Foundations can feel unmotivated	Knowledge may rest on shaky foundations

The Four Pillars of Machine Learning



Part I (this deck) lays the mathematical foundations; Part II builds the four pillars on top.

Linear Algebra

What Is Linear Algebra?

Idea

Linear algebra is the study of vectors and rules for manipulating them. Vectors are objects that can be *added together* and *multiplied by scalars*, remaining objects of the same kind.

Vectors can be surprising objects

- Geometric vectors $\mathbf{x}, \vec{y}$ (directed segments)
- Polynomials (can be added and scaled)
- Audio signals (can be added and scaled)
- Tuples of n real numbers, $\mathbf{a} \in \mathbb{R}^n$ – the focus of this book

$$\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \in \mathbb{R}^3$$

General form

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ &\vdots \qquad\qquad a_{ij}, b_i \in \mathbb{R} \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

Every n-tuple $(x_1, \ldots, x_n) \in \mathbb{R}^n$ satisfying all equations is a solution.

Three possible outcomes (Example 2.2)

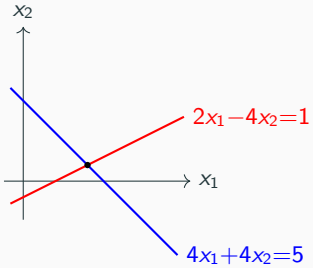
No solution	Unique solution	∞ solutions
$x_1+x_2+x_3 = 3$	$x_1+x_2+x_3 = 3$	$x_1+x_2+x_3 = 3$
$x_1-x_2+2x_3 = 2$	$x_1-x_2+2x_3 = 2$	$x_1-x_2+2x_3 = 2$
$2x_1+3x_3 = 1$	$x_2+x_3 = 2$	$2x_1+3x_3 = 5$
	$(1, 1, 1)$	

Geometric Interpretation

Each linear equation with two variables defines a line; with three variables, a plane. The solution set is the *intersection* of these lines/planes/hyperplanes.

$$4x_1 + 4x_2 = 5$$
$$2x_1 - 4x_2 = 1$$

Solution: $(x_1, x_2) = \left(1, \frac{1}{4}\right)$



$$\begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix} x_1 + \cdots + \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix} x_n = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} \iff \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

Matrices: Definition and Operations

Definition 2.1 (Matrix)

With $m, n \in \mathbb{N}$, a real-valued (m, n) matrix $\mathbf{A}$ is an $m{\cdot}n$-tuple ordered into m rows and n columns:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \qquad a_{ij} \in \mathbb{R}$$

Addition (elementwise)

$$\mathbf{A} + \mathbf{B} := [a_{ij} + b_{ij}] \in \mathbb{R}^{m \times n}$$

Multiplication $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{B} \in \mathbb{R}^{n \times k}$

$$c_{ij} = \sum_{l=1}^{n} a_{il} b_{lj}, \quad \mathbf{C} = \mathbf{AB} \in \mathbb{R}^{m \times k}$$

Caution

Matrix multiplication is *not* commutative in general: $\mathbf{AB} \neq \mathbf{BA}$. It is also *not* an elementwise (Hadamard) product.

Example 2.3

For $\mathbf{A} = \left[\begin{smallmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{smallmatrix}\right] \in \mathbb{R}^{2\times 3}$, $\mathbf{B} = \left[\begin{smallmatrix} 0 & 2 \\ 1 & -1 \\ 0 & 1 \end{smallmatrix}\right] \in \mathbb{R}^{3\times 2}$:

$$\mathbf{AB} = \begin{bmatrix} 2 & 3 \\ 2 & 5 \end{bmatrix} \in \mathbb{R}^{2\times 2}, \qquad \mathbf{BA} = \begin{bmatrix} 6 & 4 & 2 \\ -2 & 0 & 2 \\ 3 & 2 & 1 \end{bmatrix} \in \mathbb{R}^{3\times 3}$$

Identity matrix $\mathbf{I}_n \in \mathbb{R}^{n\times n}$ has 1s on the diagonal, 0 elsewhere.

Key properties ($\mathbf{A}, \mathbf{B}, \mathbf{C}$ of compatible size, $\lambda, \psi \in \mathbb{R}$)

Associativity	$(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$
Distributivity	$(\mathbf{A}+\mathbf{B})\mathbf{C} = \mathbf{AC}+\mathbf{BC}, \quad \mathbf{A}(\mathbf{C}+\mathbf{D}) = \mathbf{AC}+\mathbf{AD}$
Identity	$\mathbf{I}_m\mathbf{A} = \mathbf{A}\mathbf{I}_n = \mathbf{A}$

Definition 2.3 (Inverse)

For a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, if $\mathbf{B}$ satisfies $\mathbf{AB} = \mathbf{I}_n = \mathbf{BA}$, then $\mathbf{B} = \mathbf{A}^{-1}$ is the inverse of $\mathbf{A}$ (*regular*/*invertible*/*nonsingular*); otherwise $\mathbf{A}$ is *singular*.

Closed form for 2×2: for $\mathbf{A} = \left[\begin{smallmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{smallmatrix}\right]$,

$$\mathbf{A}^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix} \quad \text{iff } a_{11}a_{22} - a_{12}a_{21} \neq 0$$

Definition 2.4 (Transpose)

$\mathbf{B} = \mathbf{A}^\top$ with $b_{ij} = a_{ji}$.

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{I} = \mathbf{A}^{-1}\mathbf{A}, \qquad (\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

$$(\mathbf{A} + \mathbf{B})^{-1} \neq \mathbf{A}^{-1} + \mathbf{B}^{-1}$$

$$(\mathbf{A}^\top)^\top = \mathbf{A}, \qquad (\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top$$

$$(\mathbf{AB})^\top = \mathbf{B}^\top\mathbf{A}^\top$$

Definition 2.5 (Symmetric matrix)

$\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric if $\mathbf{A} = \mathbf{A}^\top$. Sum of symmetric matrices is symmetric; the *product* generally is not.

Note that only (n, n)-matrices can be symmetric – these are also called square matrices. If $\mathbf{A}$ is invertible, then so is $\mathbf{A}^\top$, and $(\mathbf{A}^{-1})^\top = (\mathbf{A}^\top)^{-1} =: \mathbf{A}^{-\top}$.

Elementary transformations (preserve the solution set)

- Exchange two equations (rows)
- Multiply an equation (row) by $\lambda \in \mathbb{R} \setminus \{0\}$
- Add two equations (rows) to each other

Definition 2.6 (Row-Echelon Form, REF)

- All-zero rows are at the bottom.
- The pivot (first nonzero entry, reading left to right) of each nonzero row is strictly right of the pivot in the row above – a "staircase" pattern.

Definition – Reduced Row-Echelon Form (RREF)

REF *and* every pivot equals 1 *and* the pivot is the only nonzero entry in its column.

Variables at pivot columns are basic variables; the rest are free variables.

Gaussian elimination

An algorithm that applies elementary transformations to bring a linear system into (reduced) row-echelon form.

Worked Example: Gaussian Elimination (2.6)

For $a \in \mathbb{R}$, solve

$$\begin{bmatrix} -2 & 4 & -2 & -1 & 4 \\ 4 & -8 & 3 & -3 & 1 \\ 1 & -2 & 1 & -1 & 1 \\ 1 & -2 & 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \\ 0 \\ a \end{bmatrix}$$

Bring the augmented matrix $[\mathbf{A} \mid \mathbf{b}]$ to row-echelon form:

$$\rightsquigarrow \cdots \rightsquigarrow \left[\begin{array}{ccccc|c} 1 & -2 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 3 & -2 \\ 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 & a+1 \end{array}\right]$$

Solvable only for $\boxed{a = -1}$. General solution (2 free parameters x_2, x_5):

$$\left\{ \mathbf{x} \in \mathbb{R}^5 : \mathbf{x} = \begin{bmatrix} 2 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix} + \lambda_1 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 2 \\ 0 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \quad \lambda_1, \lambda_2 \in \mathbb{R} \right\}$$

Recipe: Particular + General Solution

Three-step recipe

1. Find a particular solution to $\mathbf{Ax} = \mathbf{b}$.
2. Find all solutions to the homogeneous system $\mathbf{Ax} = \mathbf{0}$.
3. Combine: general solution = particular + homogeneous solutions.

The Minus-1 Trick

Given $\mathbf{A}$ in RREF, augment it to a square matrix $\bar{\mathbf{A}}$ by inserting rows of the form $[0 \cdots 0 \ -1 \ 0 \cdots 0]$ so the diagonal contains only 1 or -1. The columns of $\bar{\mathbf{A}}$ containing -1 on the diagonal are a basis of the solution space of $\mathbf{Ax} = \mathbf{0}$.

Minus-1 Trick – Worked Example (2.8)

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & -4 \end{bmatrix} \longrightarrow \bar{\mathbf{A}} = \begin{bmatrix} 1 & 3 & 0 & 0 & 3 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}$$

Reading off the columns with -1 on the diagonal directly gives a basis of $\ker(\mathbf{A})$:

$$\ker(\mathbf{A}) = \operatorname{span}\left[\begin{bmatrix} 3 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 9 \\ -4 \\ -1 \end{bmatrix}\right]$$

This bypasses solving the homogeneous system "by hand" once $\mathbf{A}$ is in reduced row-echelon form.

To find $\mathbf{A}^{-1}$, solve $\mathbf{AX} = \mathbf{I}_n$ for $\mathbf{X} = \mathbf{A}^{-1}$ using the augmented system:

$$[\mathbf{A} \,|\, \mathbf{I}_n] \rightsquigarrow \cdots \rightsquigarrow [\mathbf{I}_n \,|\, \mathbf{A}^{-1}]$$

Example 2.9

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \longrightarrow \mathbf{A}^{-1} = \begin{bmatrix} -1 & 2 & -2 & 2 \\ 1 & -1 & 2 & -2 \\ 1 & -1 & 1 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix}$$

Beyond Exact Elimination

- If $\mathbf{A}$ has full column rank: $\mathbf{A}\mathbf{x} = \mathbf{b} \iff \mathbf{x} = (\mathbf{A}^\top \mathbf{A})^{-1}\mathbf{A}^\top \mathbf{b}$, using the Moore–Penrose pseudo-inverse.
- Gaussian elimination scales *cubically* in the number of variables – impractical for systems with millions of unknowns.
- Practical alternative: iterative methods (Richardson, Jacobi, Gauß–Seidel, conjugate gradients, GMRES) of the form

$$\mathbf{x}^{(k+1)} = \mathbf{C}\mathbf{x}^{(k)} + \mathbf{d}$$

for suitable $\mathbf{C}, \mathbf{d}$, which reduce the residual error $\|\mathbf{x}^{(k+1)} - \mathbf{x}_*\|$ at every iteration and converge to a solution $\mathbf{x}_*$.

Definition 2.7 (Group)

$G := (\mathcal{G}, \otimes)$ is a group if:

1. **Closure:** $\forall x, y \in \mathcal{G} : x \otimes y \in \mathcal{G}$
2. **Associativity:** $(x \otimes y) \otimes z = x \otimes (y \otimes z)$
3. **Neutral element:** $\exists e : x \otimes e = x = e \otimes x$
4. **Inverse element:** $\forall x \; \exists y : x \otimes y = e = y \otimes x$

If additionally $x \otimes y = y \otimes x$ for all x, y, G is Abelian.

Groups: Examples (Example 2.10)

Set/operation	Closure	Neutral elt.	Inverse elt.
$(\mathbb{Z}, +)$	Y	Y	Y **(Abelian)**
$(\mathbb{N}_0, +)$	Y	Y (0)	N
$(\mathbb{Z}, \cdot)$	Y	Y (1)	N
$(\mathbb{R}, \cdot)$	Y	Y	N (0 has none)
$(\mathbb{R} \setminus \{0\}, \cdot)$	Y	Y	Y **(Abelian)**
$(\mathbb{R}^{n \times n}, \cdot)$ regular matrices	Y	$\mathbf{I}_n$	Y **(not Abelian)**

The set of invertible (regular) matrices with matrix multiplication is the general linear group $GL(n, \mathbb{R})$ – a group, but *not* Abelian since $\mathbf{AB} \neq \mathbf{BA}$ in general.

Definition 2.9 (Vector Space)

$V = (\mathcal{V}, +, \cdot)$ is a real vector space if $(\mathcal{V}, +)$ is an Abelian group and, for all $\lambda, \psi \in \mathbb{R}$, $\mathbf{x}, \mathbf{y} \in \mathcal{V}$:

- Distributivity: $\lambda \cdot (\mathbf{x} + \mathbf{y}) = \lambda\mathbf{x} + \lambda\mathbf{y}$, $(\lambda + \psi)\mathbf{x} = \lambda\mathbf{x} + \psi\mathbf{x}$
- Associativity (outer op.): $\lambda \cdot (\psi \cdot \mathbf{x}) = (\lambda\psi) \cdot \mathbf{x}$
- Neutral element: $1 \cdot \mathbf{x} = \mathbf{x}$

Example 2.11

- $V = \mathbb{R}^n$: elementwise addition and scalar multiplication.
- $V = \mathbb{R}^{m \times n}$: elementwise matrix addition and scaling.
- $V = \mathbb{C}$, with the usual addition of complex numbers.

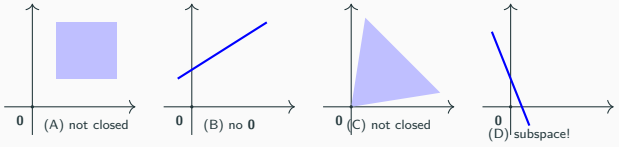
Caution – there is no "vector $\times$ vector"

Neither elementwise ("array") multiplication nor $n \times n$ matrix multiplication of two n-vectors is a defined vector-space operation. Only $\mathbf{a}\mathbf{b}^\top \in \mathbb{R}^{n \times n}$ (outer product) and $\mathbf{a}^\top\mathbf{b} \in \mathbb{R}$ (inner/dot product) are defined.

Definition 2.10 (Vector Subspace)

$U = (\mathcal{U}, +, \cdot)$ with $\emptyset \neq \mathcal{U} \subseteq \mathcal{V}$ is a subspace of V iff, restricted to $\mathcal{U}$, it is itself a vector space. It suffices to check:

$$\mathbf{0} \in \mathcal{U}, \qquad \forall \lambda \in \mathbb{R}, \mathbf{x} \in \mathcal{U} : \lambda \mathbf{x} \in \mathcal{U}, \qquad \forall \mathbf{x}, \mathbf{y} \in \mathcal{U} : \mathbf{x} + \mathbf{y} \in \mathcal{U}$$



Only line **D** (through the origin) is a subspace of $\mathbb{R}^2$.
The solution set of a *homogeneous* system $\mathbf{Ax} = \mathbf{0}$ is always a subspace; the solution set of $\mathbf{Ax} = \mathbf{b}$, $\mathbf{b} \neq \mathbf{0}$ is *not*.

Definition 2.11 (Linear Combination)

$$\mathbf{v} = \lambda_1 \mathbf{x}_1 + \cdots + \lambda_k \mathbf{x}_k = \sum_{i=1}^{k} \lambda_i \mathbf{x}_i \in V, \qquad \lambda_i \in \mathbb{R}$$

Definition 2.12 (Linear (In)dependence)

$\mathbf{x}_1, \ldots, \mathbf{x}_k$ are linearly dependent if a *non-trivial* combination gives $\mathbf{0}$:

$$\mathbf{0} = \sum_{i=1}^{k} \lambda_i \mathbf{x}_i, \qquad \text{with some } \lambda_i \neq 0$$

If only the trivial solution ($\lambda_1 = \cdots = \lambda_k = 0$) exists, they are linearly independent.

Checking Linear Independence in Practice

Practical test

Write the vectors as columns of a matrix $\mathbf{A}$ and row-reduce. **Pivot columns** $\Rightarrow$ independent vectors; any **non-pivot** column is a linear combination of pivot columns to its left, so the whole set is dependent.

Example 2.14 ($\mathbb{R}^4$)

$\begin{bmatrix} 1 \\ 2 \\ -3 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ 1 \\ 1 \end{bmatrix}$ row-reduce to a matrix with 3 pivots $\Rightarrow$ the three vectors are linearly independent.

Definition 2.13 (Generating Set / Span)

$\mathcal{A} = \{\mathbf{x}_1, \ldots, \mathbf{x}_k\} \subseteq V$ is a generating set of V if every $\mathbf{v} \in V$ is a linear combination of $\mathcal{A}$. The set of all such combinations is the span, $V = \text{span}[\mathcal{A}]$.

Definition 2.14 (Basis)

A generating set $\mathcal{A}$ is minimal if no proper subset still spans V. A minimal generating set is called a basis.

Equivalent characterizations of a basis $\mathcal{B}$

$\mathcal{B}$ is a basis $\iff$ $\mathcal{B}$ is a minimal generating set $\iff$ $\mathcal{B}$ is a maximal linearly independent set $\iff$ every $\mathbf{x} \in V$ has a *unique* representation $\mathbf{x} = \sum_i \lambda_i \mathbf{b}_i$.

Example 2.16 – several bases of $\mathbb{R}^3$

$$\mathcal{B}_{\text{std}} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}, \quad \mathcal{B}_1 = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

The dimension $\dim(V)$ = number of basis vectors – *every* basis of V has the same size.

Rank

Definition – Rank

$\text{rk}(\mathbf{A})$ = the number of linearly independent columns of $\mathbf{A} \in \mathbb{R}^{m \times n}$, which always equals the number of linearly independent rows.

Key properties

- $\text{rk}(\mathbf{A}) = \text{rk}(\mathbf{A}^\top)$ – column rank = row rank.
- Columns of $\mathbf{A}$ span a subspace $U \subseteq \mathbb{R}^m$ (the image) with $\dim(U) = \text{rk}(\mathbf{A})$.
- $\mathbf{A} \in \mathbb{R}^{n \times n}$ is regular (invertible) $\iff \text{rk}(\mathbf{A}) = n$.
- $\mathbf{Ax} = \mathbf{b}$ is solvable $\iff \text{rk}(\mathbf{A}) = \text{rk}(\mathbf{A}|\mathbf{b})$.
- The solution space of $\mathbf{Ax} = \mathbf{0}$ has dimension $n - \text{rk}(\mathbf{A})$ (the kernel).
- Full rank: $\text{rk}(\mathbf{A}) = \min(m, n)$; otherwise $\mathbf{A}$ is rank deficient.

Rank – Worked Example (2.18)

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \\ 3 & 5 & 0 \end{bmatrix} \rightsquigarrow \cdots \rightsquigarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \implies \operatorname{rk}(\mathbf{A}) = 2$$

Two linearly independent rows/columns remain after Gaussian elimination, so $\mathbf{A}$ is *not* full rank (full rank would be 3 here) – it is rank deficient.

Definition 2.15 (Linear Mapping)

$\Phi : V \to W$ preserves vector-space structure if

$$\Phi(\lambda \mathbf{x} + \psi \mathbf{y}) = \lambda \Phi(\mathbf{x}) + \psi \Phi(\mathbf{y}) \qquad \forall \mathbf{x}, \mathbf{y} \in V,\ \lambda, \psi \in \mathbb{R}$$

Definition 2.16 – special mappings $\Phi : \mathcal{V} \to \mathcal{W}$

- **Injective**: $\Phi(\mathbf{x}) = \Phi(\mathbf{y}) \Rightarrow \mathbf{x} = \mathbf{y}$
- **Surjective**: $\Phi(\mathcal{V}) = \mathcal{W}$
- **Bijective**: injective + surjective

Name	Property
Isomorphism	$\Phi : V \to W$ linear and bijective
Endomorphism	$\Phi : V \to V$ linear
Automorphism	$\Phi : V \to V$ linear and bijective

Theorem 2.17

Finite-dimensional V, W are isomorphic $\iff \dim(V) = \dim(W)$. In particular $\mathbb{R}^{m \times n} \cong \mathbb{R}^{mn}$.

For ordered bases $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ of V and $C = (\mathbf{c}_1, \dots, \mathbf{c}_m)$ of W:

$$\Phi(\mathbf{b}_j) = \alpha_{1j}\mathbf{c}_1 + \cdots + \alpha_{mj}\mathbf{c}_m = \sum_{i=1}^{m} \alpha_{ij}\mathbf{c}_i$$

Definition 2.19 (Transformation Matrix)

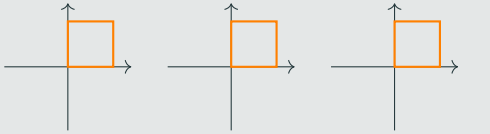
$\mathbf{A}_\Phi(i, j) = \alpha_{ij}$: the coordinates of $\Phi(\mathbf{b}_j)$ w.r.t. C form the j-th column of $\mathbf{A}_\Phi$. Then for $\hat{\mathbf{x}}$ (coords. of $\mathbf{x}$ in B) and $\hat{\mathbf{y}}$ (coords. of $\Phi(\mathbf{x})$ in C):

$$\hat{\mathbf{y}} = \mathbf{A}_\Phi \hat{\mathbf{x}}$$

This means the transformation matrix maps coordinates w.r.t. a basis in V to coordinates w.r.t. a basis in W.

Example: Linear Transformations in $\mathbb{R}^2$

Three linear transformations in $\mathbb{R}^2$ (Figure 2.10)



$$\mathbf{A}_1 = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{bmatrix}, \quad \mathbf{A}_2 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{A}_3 = \frac{1}{2}\begin{bmatrix} 3 & -1 \\ 1 & -1 \end{bmatrix}$$

Theorem 2.20 (Basis Change)

For $\Phi : V \to W$ with transformation matrix $\mathbf{A}_\Phi$ w.r.t. bases B, C, and new bases $\bar{B}, \bar{C}$, the new transformation matrix is

$$\bar{\mathbf{A}}_\Phi = \mathbf{T}^{-1}\mathbf{A}_\Phi\mathbf{S}$$

where $\mathbf{S}$ maps $\bar{B}$-coordinates to B-coordinates, and $\mathbf{T}$ maps $\bar{C}$-coordinates to C-coordinates.

Example 2.23

$\mathbf{A} = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$ w.r.t. the standard basis. With new basis $B = \left(\left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right], \left[\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right]\right)$:

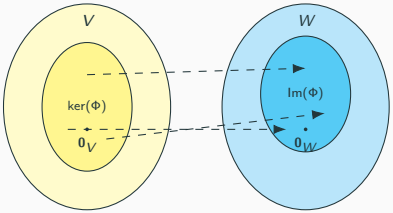
$$\bar{\mathbf{A}} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$ – a much simpler, diagonal matrix!

Definitions 2.21/2.22

$\mathbf{A}, \bar{\mathbf{A}}$ are **equivalent** if $\bar{\mathbf{A}} = \mathbf{T}^{-1}\mathbf{A}\mathbf{S}$ for regular $\mathbf{S}, \mathbf{T}$. They are **similar** (the special case $\mathbf{S} = \mathbf{T}$) if $\bar{\mathbf{A}} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}$. Similar $\Rightarrow$ equivalent, not conversely.

Definition 2.23

$$\ker(\Phi) := \Phi^{-1}(\mathbf{0}_W) = \{\mathbf{v} \in V : \Phi(\mathbf{v}) = \mathbf{0}_W\}, \qquad \operatorname{Im}(\Phi) := \{\mathbf{w} \in W \mid \exists \mathbf{v} \in V : \Phi(\mathbf{v}) = \mathbf{w}\}$$



- Φ is injective $\iff \ker(\Phi) = \{\mathbf{0}\}$
- $\operatorname{Im}(\Phi) = \operatorname{span}[\mathbf{a}_1, \ldots, \mathbf{a}_n]$, the column space of $\mathbf{A}_\Phi = [\mathbf{a}_1, \ldots, \mathbf{a}_n]$
- $\operatorname{rk}(\mathbf{A}) = \dim(\operatorname{Im}(\Phi))$
- $\ker(\Phi)$ is the general solution of $\mathbf{Ax} = \mathbf{0}$

Theorem 2.24 (Rank–Nullity Theorem)

$$\dim(\ker(\Phi)) + \dim(\operatorname{Im}(\Phi)) = \dim(V)$$

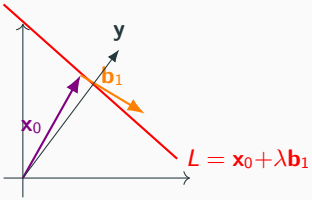
If $\dim(V) = \dim(W)$: injective $\iff$ surjective $\iff$ bijective.

Definition 2.25 (Affine Subspace)

For a vector space V, $\mathbf{x}_0 \in V$, subspace $U \subseteq V$:

$$L = \mathbf{x}_0 + U := \{\mathbf{x}_0 + \mathbf{u} : \mathbf{u} \in U\}$$

U is the direction space, $\mathbf{x}_0$ the support point. L need *not* contain $\mathbf{0}$.



Lines: $\mathbf{y} = \mathbf{x}_0 + \lambda\mathbf{b}_1$
Planes: $\mathbf{y} = \mathbf{x}_0 + \lambda_1\mathbf{b}_1 + \lambda_2\mathbf{b}_2$
Hyperplanes (dimension $n-1$): $\mathbf{y} = \mathbf{x}_0 + \sum_{i=1}^{n-1} \lambda_i\mathbf{b}_i$
The solution set of $\mathbf{Ax} = \mathbf{b}$ (if non-empty) is an affine subspace of dimension $n - \mathrm{rk}(\mathbf{A})$.

Definition 2.26 (Affine Mapping)

For linear $\Phi : V \to W$ and $\mathbf{a} \in W$:

$$\phi : V \to W, \qquad \mathbf{x} \mapsto \mathbf{a} + \Phi(\mathbf{x})$$

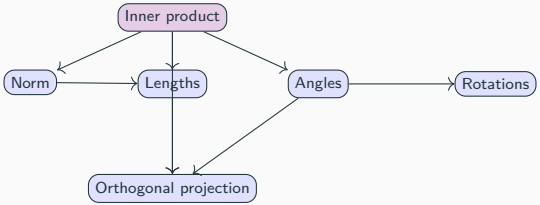
$\mathbf{a}$ is the *translation vector*. Affine maps preserve dimension and parallelism.

Analytic Geometry

Chapter 2 studied vectors, vector spaces, and linear mappings abstractly. Now we add geometric intuition: lengths, distances, and angles.

The key idea

Equip the vector space with an inner product, which induces a norm and thereby the geometry of the space.



These tools underlie orthogonal projections (Ch. 9 regression, Ch. 10 PCA), matrix decompositions (Ch. 4), and support vector machines (Ch. 12).

Definition 3.1 (Norm)

A norm on a vector space V is $\|\cdot\| : V \to \mathbb{R},\ \mathbf{x} \mapsto \|\mathbf{x}\|$, assigning each vector its length, such that for all $\lambda \in \mathbb{R},\ \mathbf{x}, \mathbf{y} \in V$:

- **Absolutely homogeneous:** $\|\lambda \mathbf{x}\| = |\lambda| \, \|\mathbf{x}\|$
- **Triangle inequality:** $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$
- **Positive definite:** $\|\mathbf{x}\| \geq 0$ and $\|\mathbf{x}\| = 0 \iff \mathbf{x} = \mathbf{0}$

Manhattan (ℓ_1) norm

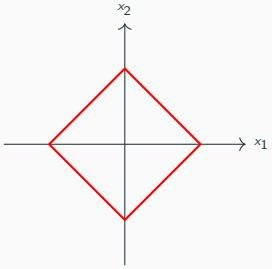
$$\|\mathbf{x}\|_1 := \sum_{i=1}^{n} |x_i|$$

Euclidean (ℓ_2) norm

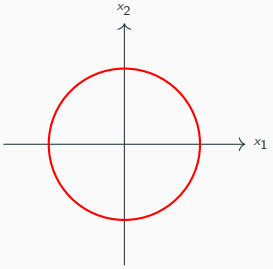
$$\|\mathbf{x}\|_2 := \sqrt{\sum_{i=1}^{n} x_i^2} = \sqrt{\mathbf{x}^\top \mathbf{x}}$$

Unit Balls of Different Norms

The set of vectors with norm 1 looks different for different norms (Figure 3.3):



$\|\mathbf{x}\|_1 = 1$ (diamond)



$\|\mathbf{x}\|_2 = 1$ (circle)

By default

Unless stated otherwise, this deck (and the book) uses the Euclidean norm $\|\cdot\|_2$.

Dot Product and Bilinear Mappings

Dot product in $\mathbb{R}^n$ (a familiar special case of an inner product):

$$\mathbf{x}^\top \mathbf{y} = \sum_{i=1}^{n} x_i y_i$$

Bilinear mapping

$\Omega : V \times V \to \mathbb{R}$ is bilinear if it is linear in *each* argument:

$$\Omega(\lambda\mathbf{x} + \psi\mathbf{y}, \mathbf{z}) = \lambda\,\Omega(\mathbf{x}, \mathbf{z}) + \psi\,\Omega(\mathbf{y}, \mathbf{z}), \qquad \Omega(\mathbf{x}, \lambda\mathbf{y} + \psi\mathbf{z}) = \lambda\,\Omega(\mathbf{x}, \mathbf{y}) + \psi\,\Omega(\mathbf{x}, \mathbf{z})$$

Definitions 3.2 – 3.3

Ω is symmetric if $\Omega(\mathbf{x}, \mathbf{y}) = \Omega(\mathbf{y}, \mathbf{x})$; positive definite if $\Omega(\mathbf{x}, \mathbf{x}) > 0$ for $\mathbf{x} \neq \mathbf{0}$ (and $\Omega(\mathbf{0}, \mathbf{0}) = 0$). A positive-definite, symmetric bilinear mapping is called an inner product, written $\langle \mathbf{x}, \mathbf{y} \rangle$. $(V, \langle \cdot, \cdot \rangle)$ with the dot product is a Euclidean vector space.

An Inner Product Need Not Be the Dot Product

Example 3.3

On $V = \mathbb{R}^2$, define

$$\langle \mathbf{x}, \mathbf{y} \rangle := x_1 y_1 - (x_1 y_2 + x_2 y_1) + 2 x_2 y_2$$

This is a valid inner product – but a *different* one from the dot product.

From basis coordinates to a matrix (Sec. 3.2.3)

For an ordered basis $B = (\mathbf{b}_1, \ldots, \mathbf{b}_n)$ and $A_{ij} := \langle \mathbf{b}_i, \mathbf{b}_j \rangle$:

$$\langle \mathbf{x}, \mathbf{y} \rangle = \hat{\mathbf{x}}^\top \mathbf{A}\, \hat{\mathbf{y}}$$

where $\hat{\mathbf{x}}, \hat{\mathbf{y}}$ are coordinate vectors w.r.t. B. Symmetry and positive definiteness of the inner product force $\mathbf{A}$ to be symmetric, positive definite.

Definition 3.4

A symmetric $\mathbf{A} \in \mathbb{R}^{n \times n}$ is positive definite if $\mathbf{x}^\top \mathbf{A}\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$ (positive semidefinite if only $\geq$ holds).

Example 3.4

$$\mathbf{A}_1 = \begin{bmatrix} 9 & 6 \\ 6 & 5 \end{bmatrix}, \qquad \mathbf{A}_2 = \begin{bmatrix} 9 & 6 \\ 6 & 3 \end{bmatrix}$$

$$\mathbf{x}^\top \mathbf{A}_1 \mathbf{x} = 9x_1^2 + 12x_1x_2 + 5x_2^2 = (3x_1 + 2x_2)^2 + x_2^2 > 0 \;\Rightarrow\; \mathbf{A}_1 \text{ pos. definite}$$

$$\mathbf{x}^\top \mathbf{A}_2 \mathbf{x} = (3x_1 + 2x_2)^2 - x_2^2 \quad \text{can be} < 0 \text{ (e.g. } \mathbf{x} = [2, -3]^\top) \;\Rightarrow\; \mathbf{A}_2 \text{ is not}$$

Theorem 3.5

$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ is an inner product $\iff$ there is a symmetric, positive definite $\mathbf{A}$ with $\langle \mathbf{x}, \mathbf{y} \rangle = \hat{\mathbf{x}}^\top \mathbf{A}\hat{\mathbf{y}}$. Then $\ker(\mathbf{A}) = \{\mathbf{0}\}$ and all diagonal entries $a_{ii} > 0$.

Every inner product induces a norm

$$\|\mathbf{x}\| := \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$$

(not every norm comes from an inner product – e.g. the Manhattan norm does not.)

Cauchy–Schwarz Inequality

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \, \|\mathbf{y}\|$$

Definition 3.6 (Distance / Metric)

$$d(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\| = \sqrt{\langle \mathbf{x} - \mathbf{y}, \mathbf{x} - \mathbf{y} \rangle}$$

A metric d is positive definite, symmetric, and satisfies the triangle inequality $d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})$.

Contrast

Inner products and metrics behave *oppositely*: similar $\mathbf{x}, \mathbf{y}$ give a *large* inner product but a *small* distance.

Cauchy–Schwarz guarantees $-1 \le \dfrac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\|\|\mathbf{y}\|} \le 1$, so there is a unique $\omega \in [0, \pi]$ with

$$\cos\omega = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\|\,\|\mathbf{y}\|}$$

Example 3.6

For $\mathbf{x} = [1,1]^\top$, $\mathbf{y} = [1,2]^\top$ (dot product):

$$\cos\omega = \frac{3}{\sqrt{10}} \Rightarrow \omega \approx 0.32 \text{ rad} \approx 18^\circ$$

Definition 3.7 (Orthogonality)

$\mathbf{x} \perp \mathbf{y} \iff \langle \mathbf{x}, \mathbf{y} \rangle = 0$. If additionally $\|\mathbf{x}\| = \|\mathbf{y}\| = 1$: orthonormal.

Orthogonality depends on the inner product!

$\mathbf{x} = [1,1]^\top$, $\mathbf{y} = [-1,1]^\top$ are orthogonal under the dot product ($\omega = 90^\circ$), but with $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \left[\begin{smallmatrix} 2 & 0 \\ 0 & 1 \end{smallmatrix}\right] \mathbf{y}$ we get $\cos\omega = -\frac{1}{3}$, i.e. $\omega \approx 109.5^\circ$: *not* orthogonal.

Definition 3.8

A square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is orthogonal iff its columns are orthonormal, so that

$$\mathbf{A}\mathbf{A}^\top = \mathbf{I} = \mathbf{A}^\top \mathbf{A} \qquad \Longrightarrow \qquad \mathbf{A}^{-1} = \mathbf{A}^\top$$

Why orthogonal matrices matter

Transformations by an orthogonal matrix preserve both length and angle:

$$\|\mathbf{A}\mathbf{x}\|^2 = \mathbf{x}^\top \mathbf{A}^\top \mathbf{A}\mathbf{x} = \mathbf{x}^\top \mathbf{x} = \|\mathbf{x}\|^2$$

$$\cos\omega = \frac{(\mathbf{A}\mathbf{x})^\top (\mathbf{A}\mathbf{y})}{\|\mathbf{A}\mathbf{x}\|\,\|\mathbf{A}\mathbf{y}\|} = \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\|\,\|\mathbf{y}\|}$$

Orthogonal matrices define rotations (possibly combined with flips) – see the end of this chapter.

Orthonormal Basis

Definition 3.9 (ONB)

A basis $\{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ of an n-dimensional V is

$$\langle \mathbf{b}_i, \mathbf{b}_j \rangle = 0 \quad (i \neq j) \quad \text{(orthogonal basis)}, \qquad \langle \mathbf{b}_i, \mathbf{b}_i \rangle = 1 \quad \text{(orthonormal, ONB)}$$

Example 3.8

The canonical basis of $\mathbb{R}^n$ is an ONB under the dot product. Also, in $\mathbb{R}^2$:

$$\mathbf{b}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \mathbf{b}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

form an ONB since $\mathbf{b}_1^\top \mathbf{b}_2 = 0$ and $\|\mathbf{b}_1\| = 1 = \|\mathbf{b}_2\|$.

Constructing an ONB

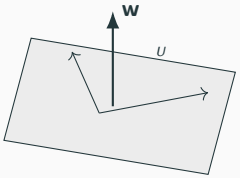
Any basis can be turned into an ONB via Gram-Schmidt orthogonalization (detailed later this chapter).

Orthogonal Complement

Definition

For a D-dimensional V and M-dimensional subspace $U \subseteq V$, the orthogonal complement $U^\perp$ is the $(D-M)$-dimensional subspace containing all vectors orthogonal to every vector in U. $U \cap U^\perp = \{\mathbf{0}\}$, and any $\mathbf{x} \in V$ decomposes uniquely as

$$\mathbf{x} = \sum_{m=1}^{M} \lambda_m \mathbf{b}_m + \sum_{j=1}^{D-M} \psi_j \mathbf{b}_j^\perp$$



In $\mathbb{R}^3$, the orthogonal complement of a plane U is spanned by a single normal vector $\mathbf{w}$, $\|\mathbf{w}\| = 1$, orthogonal to U. Orthogonal complements generalize to describe hyperplanes in n-dimensional spaces.

The idea of an inner product generalizes beyond finite vectors: a sum over components becomes an integral.

Definition

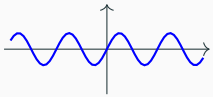
For $u, v : \mathbb{R} \to \mathbb{R}$,

$$\langle u, v \rangle := \int_a^b u(x)v(x)\,dx$$

If this evaluates to 0, u and v are orthogonal functions.

Example 3.9

For $u = \sin(x)$, $v = \cos(x)$ on $[-\pi, \pi]$:
$f(x) = u(x)v(x)$ is *odd*, so $\int_{-\pi}^{\pi} f(x)\,dx = 0$. Hence sin and cos are orthogonal.



A larger orthogonal family

$\{1, \cos(x), \cos(2x), \dots\}$ is a mutually orthogonal collection of functions on $[-\pi, \pi]$: this is the foundation of Fourier series – projecting functions onto the subspace they span.

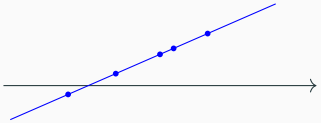
Just as with finite vectors, orthogonality of functions lets us decompose complex signals into simple, independent components – a recurring theme throughout the rest of the book (e.g. Gaussian processes, Sec. 6.4.6).

Why Project?

High-dimensional data is hard to analyze, but often most of the information lives in only a few dimensions. Orthogonal projections let us compress data onto an informative lower-dimensional subspace with minimal loss.

Definition 3.10 (Projection)

$\pi : V \to U$ is a projection if $\pi^2 = \pi \circ \pi = \pi$ (idempotent). The corresponding transformation matrix $\mathbf{P}_\pi$ satisfies $\mathbf{P}_\pi^2 = \mathbf{P}_\pi$.



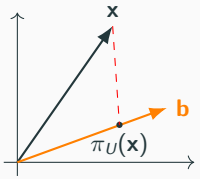
Orthogonal projection of points onto a line

Uses: PCA (Ch. 10), linear regression (Ch. 9), SVM classification (Ch. 12).

Projection onto a Line (1D Subspace) – Setup

Let $U = \text{span}[\mathbf{b}] \subseteq \mathbb{R}^n$ be a line through the origin. We seek $\pi_U(\mathbf{x}) \in U$ closest to $\mathbf{x}$.

- $\pi_U(\mathbf{x})$ minimizes $\|\mathbf{x} - \pi_U(\mathbf{x})\|$, so the displacement $\pi_U(\mathbf{x}) - \mathbf{x}$ is orthogonal to $\mathbf{b}$.
- $\pi_U(\mathbf{x}) \in U$, so $\pi_U(\mathbf{x}) = \lambda\mathbf{b}$ for some $\lambda \in \mathbb{R}$.



Orthogonality condition

$$\langle \mathbf{x} - \pi_U(\mathbf{x}), \mathbf{b} \rangle = 0 \quad \overset{\pi_U(\mathbf{x}) = \lambda\mathbf{b}}{\Longleftrightarrow} \quad \langle \mathbf{x} - \lambda\mathbf{b}, \mathbf{b} \rangle = 0$$

Projection onto a Line – Solving for λ, $\pi_U(\mathbf{x})$, $\mathbf{P}_\pi$

Exploiting bilinearity and symmetry of the inner product:

$$\lambda = \frac{\langle \mathbf{x}, \mathbf{b} \rangle}{\langle \mathbf{b}, \mathbf{b} \rangle} = \frac{\mathbf{b}^\top \mathbf{x}}{\|\mathbf{b}\|^2} \qquad \text{(dot product)}$$

$$\pi_U(\mathbf{x}) = \lambda \mathbf{b} = \frac{\mathbf{b}^\top \mathbf{x}}{\|\mathbf{b}\|^2} \mathbf{b}, \qquad \|\pi_U(\mathbf{x})\| = |\lambda| \, \|\mathbf{b}\| = |\cos \omega| \, \|\mathbf{x}\|$$

Projection matrix

Since $\pi_U(\mathbf{x}) = \mathbf{P}_\pi \mathbf{x}$ is linear:

$$\mathbf{P}_\pi = \frac{\mathbf{b}\mathbf{b}^\top}{\|\mathbf{b}\|^2}$$

a symmetric, rank-1 matrix. $\mathbf{P}_\pi$ maps any $\mathbf{x}$ onto the line through the origin spanned by $\mathbf{b}$.

Example 3.10

For $\mathbf{b} = [1, 2, 2]^\top$: $\mathbf{P}_\pi = \frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}$. For $\mathbf{x} = [1, 1, 1]^\top$: $\pi_U(\mathbf{x}) = \frac{1}{9}[5, 10, 10]^\top$.

Projection onto a General Subspace

Now let $U \subseteq \mathbb{R}^n$ have basis $(\mathbf{b}_1, \ldots, \mathbf{b}_m)$, $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_m]$. Write $\pi_U(\mathbf{x}) = \mathbf{B}\boldsymbol{\lambda}$.

Orthogonality conditions (m simultaneous equations)

$$\mathbf{b}_1^\top(\mathbf{x} - \mathbf{B}\boldsymbol{\lambda}) = 0, \quad \ldots, \quad \mathbf{b}_m^\top(\mathbf{x} - \mathbf{B}\boldsymbol{\lambda}) = 0 \iff \mathbf{B}^\top(\mathbf{x} - \mathbf{B}\boldsymbol{\lambda}) = \mathbf{0}$$

Normal equation

$$\mathbf{B}^\top\mathbf{B}\,\boldsymbol{\lambda} = \mathbf{B}^\top\mathbf{x} \qquad \Longrightarrow \qquad \boldsymbol{\lambda} = (\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top\mathbf{x}$$

($\mathbf{B}^\top\mathbf{B}$ is invertible since $\mathbf{b}_1, \ldots, \mathbf{b}_m$ are linearly independent.)

$$\pi_U(\mathbf{x}) = \mathbf{B}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top\mathbf{x}, \qquad \boxed{\mathbf{P}_\pi = \mathbf{B}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top}$$

$(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top$ is the Moore–Penrose pseudo-inverse of $\mathbf{B}$ – it reduces to the 1D formula when $m = 1$.

Worked Example: Projection onto a Plane in $\mathbb{R}^3$

Example 3.11

$U = \text{span}\left[\begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}0\\1\\2\end{bmatrix}\right] \subseteq \mathbb{R}^3, \quad \mathbf{x} = \begin{bmatrix}6\\0\\0\end{bmatrix}$

$$\mathbf{B} = \begin{bmatrix}1 & 0\\1 & 1\\1 & 2\end{bmatrix}, \quad \mathbf{B}^\top\mathbf{B} = \begin{bmatrix}3 & 3\\3 & 5\end{bmatrix}, \quad \mathbf{B}^\top\mathbf{x} = \begin{bmatrix}6\\0\end{bmatrix}$$

Normal equation $\Rightarrow \boldsymbol{\lambda} = [5, -3]^\top$, so

$$\pi_U(\mathbf{x}) = \mathbf{B}\boldsymbol{\lambda} = \begin{bmatrix}5\\2\\-1\end{bmatrix}, \qquad \mathbf{P}_\pi = \mathbf{B}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top = \frac{1}{6}\begin{bmatrix}5 & 2 & -1\\2 & 2 & 2\\-1 & 2 & 5\end{bmatrix}$$

Projection error: $\|\mathbf{x} - \pi_U(\mathbf{x})\| = \|[1, -2, 1]^\top\| = \sqrt{6}$.

Idea

Turn any basis $(\mathbf{b}_1, \dots, \mathbf{b}_n)$ into an orthogonal basis $(\mathbf{u}_1, \dots, \mathbf{u}_n)$ (with $\operatorname{span}[\mathbf{b}_1, \dots] = \operatorname{span}[\mathbf{u}_1, \dots]$) by iteratively projecting away what is already spanned:

$$\mathbf{u}_1 := \mathbf{b}_1, \qquad \mathbf{u}_k := \mathbf{b}_k - \pi_{\operatorname{span}[\mathbf{u}_1, \dots, \mathbf{u}_{k-1}]}(\mathbf{b}_k), \quad k = 2, \dots, n$$

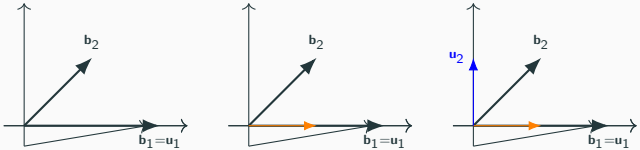
Normalizing each $\mathbf{u}_k$ (dividing by its norm) yields an orthonormal basis.

Example 3.12

$\mathbf{b}_1 = [2, 0]^\top$, $\mathbf{b}_2 = [1, 1]^\top$ in $\mathbb{R}^2$:

$$\mathbf{u}_1 = \mathbf{b}_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \qquad \mathbf{u}_2 = \mathbf{b}_2 - \frac{\mathbf{u}_1 \mathbf{u}_1^\top}{\|\mathbf{u}_1\|^2} \mathbf{b}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Gram–Schmidt – Geometric View (Figure 3.12)

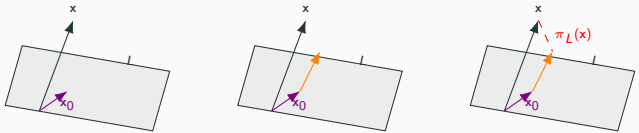


(a) original (b) project $\mathbf{b}_2$ (c) orthogonal basis

Projection onto Affine Subspaces

For an affine space $L = \mathbf{x}_0 + U$, reduce to the vector-subspace case by shifting:

$$\pi_L(\mathbf{x}) = \mathbf{x}_0 + \pi_U(\mathbf{x} - \mathbf{x}_0)$$



(a) setting (b) shift by $-\mathbf{x}_0$ (c) shift back

The distance from $\mathbf{x}$ to L equals the distance from $\mathbf{x} - \mathbf{x}_0$ to U:

$$d(\mathbf{x}, L) = d(\mathbf{x} - \mathbf{x}_0,\ U)$$

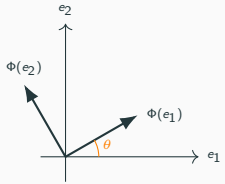
This construction underlies the separating hyperplane used in SVM classification (Ch. 12).

A rotation is a linear mapping that rotates a plane by angle θ about the origin (positive θ: counterclockwise).

Rotating the standard basis $\mathbf{e}_1, \mathbf{e}_2$ by θ:

$$\Phi(\mathbf{e}_1) = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}, \quad \Phi(\mathbf{e}_2) = \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$$

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$



Note

$\mathbf{R}(\theta)$ is an orthogonal matrix: $\mathbf{R}(\theta)^{-1} = \mathbf{R}(\theta)^\top$, and it preserves lengths and angles.

In $\mathbb{R}^3$, rotations occur about one of the three standard axes ("counterclockwise" = looking from the axis tip toward the origin):

$$\mathbf{R}_1(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}, \quad \mathbf{R}_2(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}, \quad \mathbf{R}_3(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Definition 3.11 (Givens Rotation, n dimensions)

Fix $n-2$ dimensions and rotate a 2D plane; $\mathbf{R}_{ij}(\theta) \in \mathbb{R}^{n\times n}$ is the identity matrix except

$$r_{ii} = \cos\theta, \quad r_{ij} = -\sin\theta, \quad r_{ji} = \sin\theta, \quad r_{jj} = \cos\theta, \qquad 1 \le i < j \le n$$

Properties of Rotations

- **Distance preserving:** $\|\mathbf{x} - \mathbf{y}\| = \|\mathbf{R}_\theta(\mathbf{x}) - \mathbf{R}_\theta(\mathbf{y})\|$
- **Angle preserving:** the angle between $\mathbf{R}_\theta\mathbf{x}$ and $\mathbf{R}_\theta\mathbf{y}$ equals the angle between $\mathbf{x}, \mathbf{y}$
- **Not commutative in ≥ 3 dimensions:** order matters, unless rotating about the same axis/point in $\mathbb{R}^2$, where $\mathbf{R}(\phi)\mathbf{R}(\theta) = \mathbf{R}(\theta)\mathbf{R}(\phi)$ for all $\phi, \theta \in [0, 2\pi)$ (an Abelian group under composition)

Applications

Computer graphics, robotics (orienting joints of a robotic arm), and 3D vision all rely on rotation matrices.

Summary

Recap: The Toolbox

Concept	What it gives us
Systems of linear equations	Modeling constraints; solved via Gaussian elimination
Vector spaces & subspaces	The structured world vectors live in
Linear independence, basis, rank	Minimal, non-redundant descriptions of a space
Linear & affine mappings	Structure-preserving transformations; matrices
Norms & inner products	Length, distance, similarity
Angles & orthogonality	Comparing directions; orthonormal bases
Orthogonal projections	Optimal lower-dimensional approximations
Rotations	Distance- and angle-preserving transformations

Where this leads (Part II)

Regression (Ch. 9), Dimensionality Reduction / PCA (Ch. 10), Density Estimation (Ch. 11), Classification / SVMs (Ch. 12) – all built directly on these foundations.

Further Reading

- Linear algebra: Strang (2003), Golan (2007), Axler (2015), Liesen & Mehrmann (2015).
- Numerical linear algebra / iterative solvers: Stoer & Burlirsch (2002), Golub & Van Loan (2012), Horn & Johnson (2013).
- Analytic geometry & inner product spaces: Axler (2015), Boyd & Vandenberghe (2018).
- Applications previewed here: Schölkopf & Smola (2002) [kernel methods], Rasmussen & Williams (2006) [Gaussian processes], Pearson (1901); Hotelling (1933) [PCA].

Source

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